THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Algebra Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 2	$9(a-b)=9d$
2	**(c)** 7	$(x+1/x)^2=x^2+2+1/x^2$
3	**(b)** 7	$4d+16$ = sum
4	**(c)** 18	$a^2-b^2=(a-b)(a+b)$
5	**(a)** 9	$a^2-(a-3)(a+3)=9$
6	**(a)** 22	Middle integer is 21
7	**(a)** 198	$(100a+10b+c)-(100c+10b+a)=99(a-c)$
8	**(b)** 4	$9(a-b)=9d$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$2n$ is a multiple of 2 and $2n+1=2n+1$ leaves remainder 1; R is the definition behind the expressions.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$3(4)+5=17$ so A is true; R is also true but it does not explain the value.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$a+b=9$, $a-b=3$ (1); $a=6$, $b=3$, number 63 (1).	1+1
12	$x^2-y^2=(x+y)(x-y)$ (1) $=85$ (1).	1+1
13	Identity $(n+3)^2-(n-3)^2=12n$ (1); $=12\times5=60$ (1).	1+1
14	$a+b=11$, $a-b=5$ (1); $a=8$, $b=3$, number 83 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$11(a+b)=110$ so $a+b=10$ (1); $a-b=4$ gives $a=7$, $b=3$ (1); number $=73$ (1).	1+1+1
16	$x^2+1/x^2=3^2-2=7$ (1); $x^4+1/x^4=(7)^2-2=47$ (2).	1+2
17	$4d+16=36$ gives $d=5$ (1); dates 5, 6, 12, 13 (1); sum $=4d+16$ for any d (1).	1+1+1
	OR: $11(a+b)=132$ so $a+b=12$ (1); $a-b=2$ gives $a=7$, $b=5$ (1); number $=75$ (1).	1+1+1
18	$x^2+1/x^2=4^2-2=14$ (1); $x^4+1/x^4=(14)^2-2=194$ (2).	1+2
	OR: $4d+16=52$ gives $d=9$ (1); dates 9, 10, 16, 17 (1); sum $=4d+16$ for any d (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $6n$ (1). (ii) $6n+12$ (1). (iii) $\frac{6n+12}{3}=2n+4$ (1). (iv) Subtract $2n$: $2n+4-2n=4$, independent of n (1); for $n=7$: $42, 54, 18, 18-14=4$ (1).	1+1+1+2
20	Using $(p+q)^2-(p-q)^2=4pq$ (2); $4\times3n\times2=24n$ (1); for $n=5$: 120 (2).	2+1+2
	OR: $(100a+10b+c)-(100c+10b+a)=99a-99c$ (2); $=99(a-c)=99\times2$ (1); difference $=198$ (2).	2+1+2

Mistakes to avoid:

- Adding unlike terms: $3x+4x^2=7x^3$; only like terms are combined, and $3x+4x=7x$ (not $7x^2$).
- Writing the two-digit number as ab (product) instead of $10a+b$.
- Checking a trick or identity with one example only and calling it a proof; use a variable for the general case.
- Sign mistakes when removing brackets: $-(x-3)=-x+3$, not $-x-3$.
- Writing $(a+b)^2=a^2+b^2$ or $(a-b)^2=a^2-b^2$ while simplifying expressions.
- Deriving a pattern rule from the first term only; check that the rule works for every given term (e.g. $3n+1$ for $n=1,2,3$).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Algebra Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 25	$a^2 - b^2 = (a-b)(a+b)$
2	**(b)** 9	$a^2 - (a-3)(a+3) = 9$
3	**(d)** 35	Middle integer is 34
4	**(a)** 297	$(100a + 10b + c) - (100c + 10b + a) = 99(a-c)$
5	**(a)** 5	$9(a-b) = 9d$
6	**(a)** 23	$(x + 1/x)^2 = x^2 + 2 + 1/x^2$
7	**(c)** 18	$4d + 16$ = sum
8	**(d)** 30	$a^2 - b^2 = (a-b)(a+b)$
9	**(c)** A is true but R is false.	$n + (n+1) + (n+2) = 3n + 3 = 3(n+1)$ so A is true; $2 + 3 + 4 = 9$ is not divisible by 6, so R is false.
10	**(d)** A is false but R is true.	$3x + 4x = 7x$, so A is false; R is the correct rule.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$x^2 - y^2 = (x+y)(x-y)$ (1) $= 161$ (1).	1+1
12	Identity $(n+3)^2 - (n-3)^2 = 12n$ (1); $= 12 \times 8 = 96$ (1).	1+1
13	$a + b = 13$, $a - b = 1$ (1); $a = 7$, $b = 6$, number 76 (1).	1+1
14	$x^2 - y^2 = (x+y)(x-y)$ (1) $= 124$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$11(a+b) = 154$ so $a + b = 14$ (1); $a - b = 2$ gives $a = 8$, $b = 6$ (1); number $= 86$ (1).	1+1+1
16	$x^2 + 1/x^2 = 5^2 - 2 = 23$ (1); $x^4 + 1/x^4 = (23)^2 - 2 = 527$ (2).	1+2
17	$4d + 16 = 68$ gives $d = 13$ (1); dates 13, 14, 20, 21 (1); sum $= 4d + 16$ for any d (1).	1+1+1
	OR: $11(a+b) = 88$ so $a + b = 8$ (1); $a - b = 2$ gives $a = 5$, $b = 3$ (1); number $= 53$ (1).	1+1+1
18	$x^2 + 1/x^2 = 6^2 - 2 = 34$ (1); $x^4 + 1/x^4 = (34)^2 - 2 = 1154$ (2).	1+2
	OR: $4d + 16 = 80$ gives $d = 16$ (1); dates 16, 17, 23, 24 (1); sum $= 4d + 16$ for any d (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 13 (1). (ii) $3n + 1$ (1). (iii) $3(20) + 1 = 61$ (1). (iv) $3n + 1 = 100$ gives $3n = 99$, $n = 33$ squares (1+1).	1+1+1+2
20	Using $(p+q)^2 - (p-q)^2 = 4pq$ (2); $4 \times 5n \times 2 = 40n$ (1); for $n = 3$: 120 (2).	2+1+2
	OR: $(100a + 10b + c) - (100c + 10b + a) = 99a - 99c$ (2); $= 99(a-c) = 99 \times 4$ (1); difference $= 396$ (2).	2+1+2

Mistakes to avoid:

- Adding unlike terms: $3x + 4x^2 = 7x^3$; only like terms are combined, and $3x + 4x = 7x$ (not $7x^2$).
- Writing the two-digit number as ab (product) instead of $10a + b$.
- Checking a trick or identity with one example only and calling it a proof; use a variable for the general case.
- Sign mistakes when removing brackets: $-(x-3) = -x + 3$, not $-x - 3$.
- Writing $(a+b)^2 = a^2 + b^2$ or $(a-b)^2 = a^2 - b^2$ while simplifying expressions.
- Deriving a pattern rule from the first term only; check that the rule works for every given term (e.g. $3n + 1$ for $n = 1, 2, 3$).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Algebra Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 396	$(100a+10b+c)-(100c+10b+a)=99(a-c)$
2	**(b)** 6	$9(a-b)=9d$
3	**(d)** 34	$(x+1/x)^2=x^2+2+1/x^2$
4	**(d)** 4	$4d+16$ = sum
5	**(d)** 23	$a^2-b^2=(a-b)(a+b)$
6	**(d)** 9	$a^2-(a-3)(a+3)=9$
7	**(c)** 68	Middle integer is 67
8	**(b)** 495	$(100a+10b+c)-(100c+10b+a)=99(a-c)$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$(n+1)^2-n^2=(2n+1)(1)=2n+1$ by the identity in R.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	A is true by place value; R is true since $(10a+b)+(10b+a)=11(a+b)$, but R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Identity $(n+3)^2-(n-3)^2=12n$ (1); $=12\times13=156$ (1).	1+1
12	$a+b=8$, $a-b=4$ (1); $a=6$, $b=2$, number 62 (1).	1+1
13	$x^2-y^2=(x+y)(x-y)$ (1) $=270$ (1).	1+1
14	Identity $(n+3)^2-(n-3)^2=12n$ (1); $=12\times20=240$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$11(a+b)=176$ so $a+b=16$ (1); $a-b=2$ gives $a=9$, $b=7$ (1); number $=97$ (1).	1+1+1
16	$x^2+1/x^2=7^2-2=47$ (1); $x^4+1/x^4=(47)^2-2=2207$ (2).	1+2
17	$4d+16=100$ gives $d=21$ (1); dates 21, 22, 28, 29 (1); sum $=4d+16$ for any d (1).	1+1+1
	OR: $11(a+b)=99$ so $a+b=9$ (1); $a-b=3$ gives $a=6$, $b=3$ (1); number $=63$ (1).	1+1+1
18	$x^2+1/x^2=8^2-2=62$ (1); $x^4+1/x^4=(62)^2-2=3842$ (2).	1+2
	OR: $4d+16=48$ gives $d=8$ (1); dates 8, 9, 15, 16 (1); sum $=4d+16$ for any d (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $10a+b$ (1). (ii) $10b+a$ (1). (iii) $11a+11b=11(a+b)$ (1). (iv) The sum has the factor 11 for every a, b (1); for 47: $47+74=121=11\times11$ (1).	1+1+1+2
20	Using $(p+q)^2-(p-q)^2=4pq$ (2); $4\times6n\times1=24n$ (1); for $n=2$: 48 (2).	2+1+2
	OR: $(100a+10b+c)-(100c+10b+a)=99a-99c$ (2); $=99(a-c)=99\times6$ (1); difference $=594$ (2).	2+1+2

Mistakes to avoid:

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- Writing the two-digit number as ab (product) instead of $10a+b$.
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- Sign mistakes when removing brackets: $-(x-3)=-x+3$, not $-x-3$.
- Writing $(a+b)^2=a^2+b^2$ or $(a-b)^2=a^2-b^2$ while simplifying expressions.
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